

# Variable Selection Methods

For Regression and Classification problems

## Feature Preparation/Processing

### Feature engineering

*Covariates creation and Feature transformation*

#### Covariates creation

- *Taking lags of MEVs*
- *Taking interaction terms*
- *Debt to income ratio*
- *Loan to value ratio*
- *Utilization ratio*

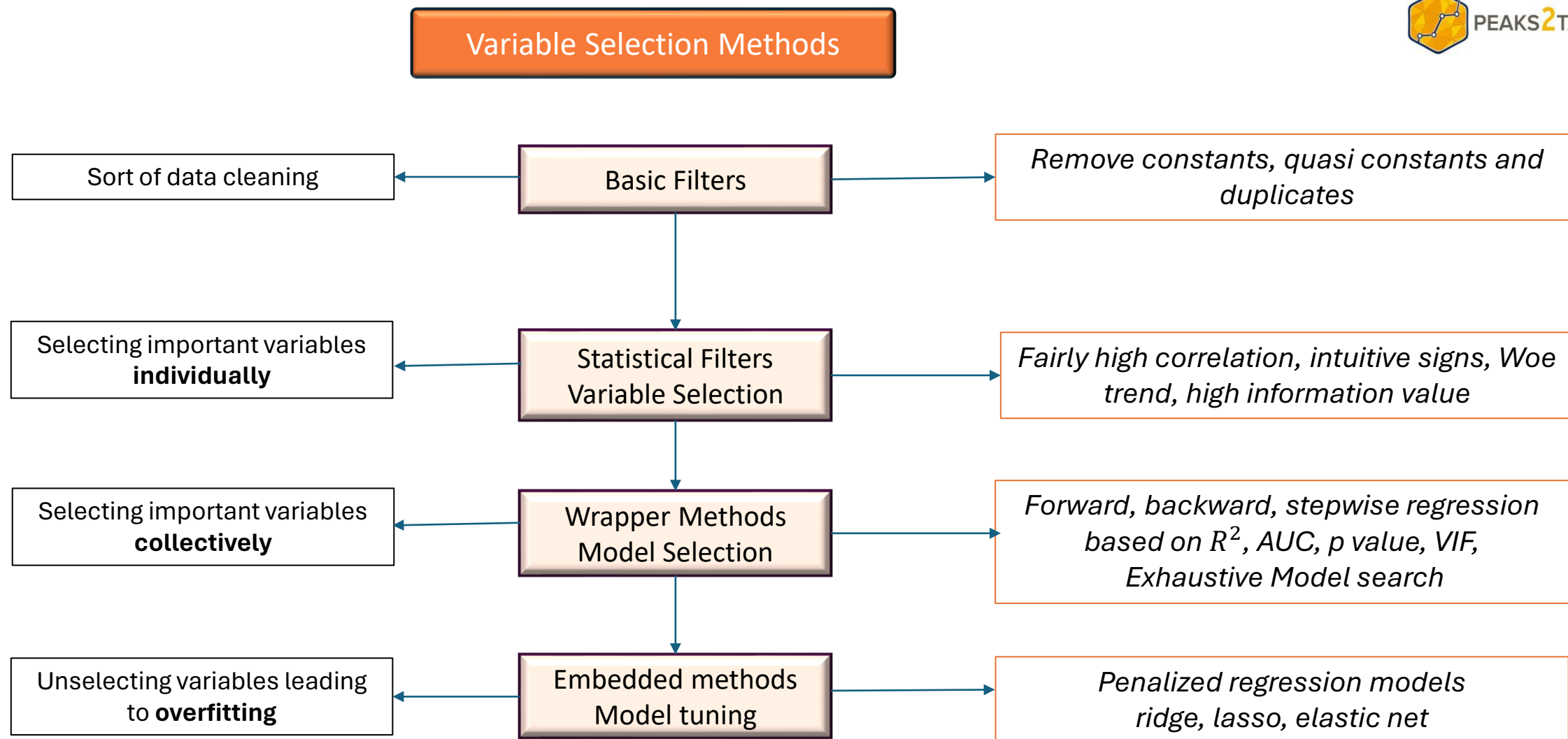
#### Features transformation

- *Lag transformation*
- *Growth rates*
- *Weight of evidence (discretization)*

### Feature selection

*Selecting important variables*

- *Basic Filters*
- *Statistical Filters*
- *Wrapper Methods*
- *Embedded Methods*



|                            | Linear Regression                                             | Logistic Regression                                                         |
|----------------------------|---------------------------------------------------------------|-----------------------------------------------------------------------------|
| 1. Co-variates creation    | Ratios, interactions                                          | Ratios, interactions<br>LTV ratio, DTI ratio, Utilisation ratio             |
| 2. Variable transformation | Lagged transformations, relative change                       | Weight of evidence transformations                                          |
| 3. Basic Filters           | Constants, quasi constants and duplicates                     | Constants, quasi constants and duplicates                                   |
| 4. Statistical Filters     | Correlation, Multicollinearity, Sign intuitiveness            | WOE trend, Information value, Gini, Chi square test, Mutual information     |
| 5. Wrapper methods         | Forward/Backward/Stepwise Regression, Exhaustive Model Search | Forward/Backward/Stepwise Regression, Exhaustive Model Search               |
| 6. Embedded Methods        | Penalised Regression – Lasso, Ridge, Elastic Net              | Penalised Regression – Lasso, Ridge, Elastic Net<br>Cost Sensitive Learning |

| Particular    | Formula                                   |
|---------------|-------------------------------------------|
| 1. Growth     | $\frac{X_t - X_{t-1}}{X_t}$               |
| 2. Difference | $X_t - X_{t-1}$                           |
| 3. MA         | Average of n quarters                     |
| 4. QoQ        | $X_t$ vs $X_{t-1}$                        |
| 5. YoY        | $X_t$ vs $X_{t-4}$                        |
| 6. Lag        | $X_{t-1}$                                 |
| 7. Leading    | $X_{t+1}$                                 |
| 8. Log Odds   | $\text{Log} \frac{DR}{1-DR}$              |
| 9. Vasicek    | Z calculated by minimizing squared errors |

| Basic Filters       | Rationale                                                                                                                       |
|---------------------|---------------------------------------------------------------------------------------------------------------------------------|
| 1. Constants        | We want to explain variation in Y through variations in X. If X has no variation, it is not useful to explain the variance in Y |
| 2. Quasi- Constants | Variables with low variance are dropped for the same reasons as above                                                           |
| 3. Duplicates       | Adding duplicate variables causing redundancy in the model. It can also lead to problems like multicollinearity                 |

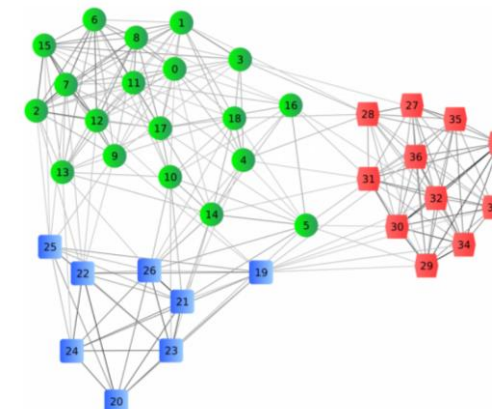
| Statistical Filters                                       | Linear Regression                                     | Logistic Regression                         |
|-----------------------------------------------------------|-------------------------------------------------------|---------------------------------------------|
| 1. Magnitude of relationship with Y should be significant | Correlation should be greater than 30%                | IV should be high. Gini should be above 10% |
| 2. Direction of relationship should be as expected        | Actual sign of correlation should be same as expected | Trend of WOE should be same as expected     |

# Variable Clustering (for dimension reduction)

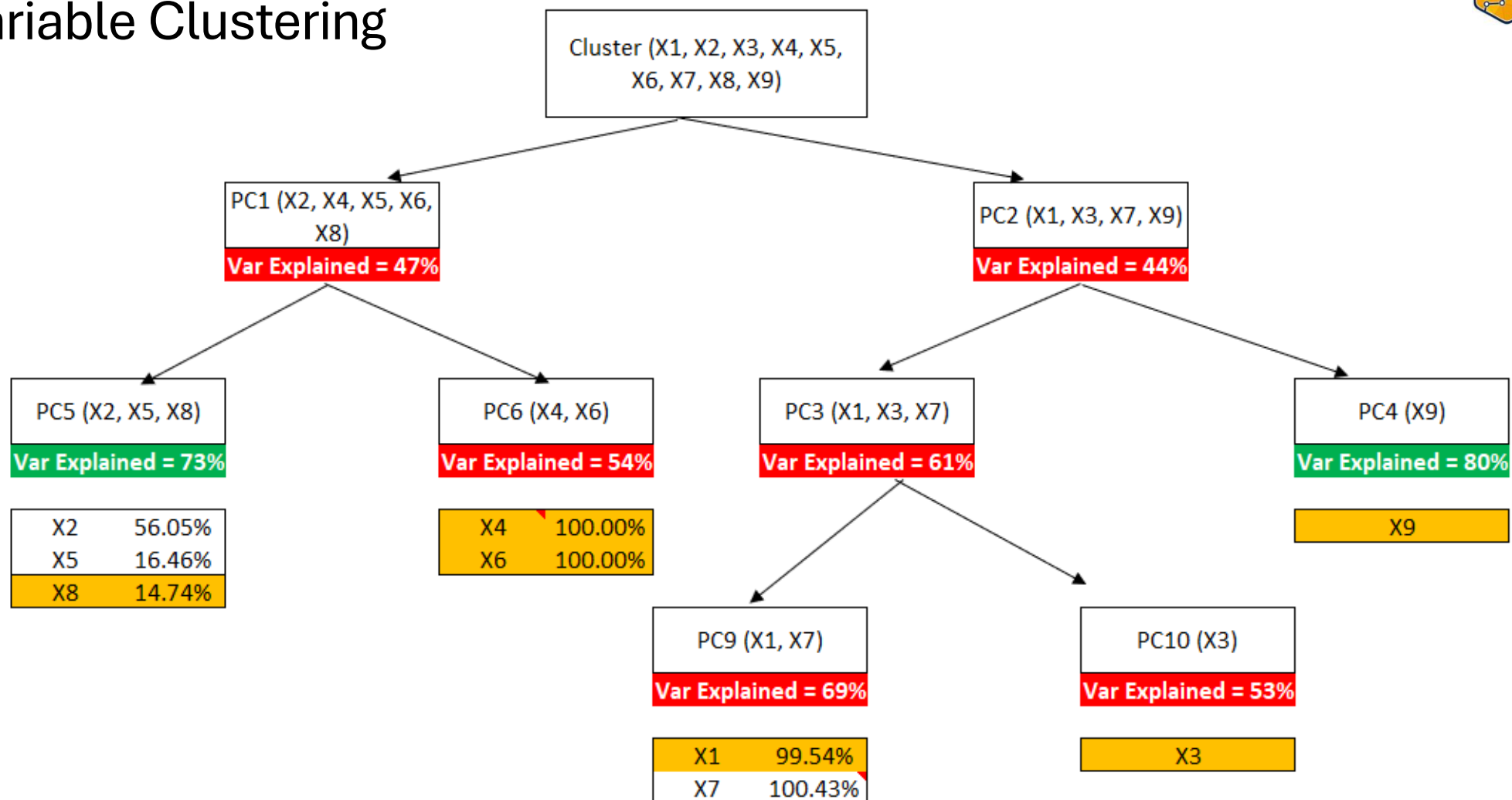
- The variable clustering (PROC VARCLUS in SAS) procedure is standard and widely used in the industry for variable selection. The VARCLUS procedure divides a set of numeric variables into either disjointed or hierarchical clusters. Associated with each cluster is a linear combination of the variables in the cluster, which may be either the first principal component or the centroid component. PROC VARCLUS displays the  $R^2$  value of each variable within its own cluster and against its nearest cluster. The lower the ratio of  $(1 - R^2_{\text{own}}) / (1 - R^2_{\text{nearest}})$  for each variable, the better it can represent the cluster. Either the top 10 variables from each cluster or ratio below some cut-off (0.5) to account for significant set of variables are selected. The cut-off used would change iteratively as the total number of variables that would appear fit to be imputed in the step-wise regression (after variable reduction techniques like correlation), regression results are examined, analyzing model performance and goodness of fit results. This technique helps in providing the first line of defense to multi-collinearity.

In simple language it's a 4-step process

- Create Principal components** – Think of these as clusters
- Assign variables** to Principal components – Based on correlation of each variable to each component
- Create Subcomponents** – If the variables in cluster don't have significant correlation the principal component, then go more granular
- Choose the best variable** – Out of all the variables that belong to a cluster, choose a variable which has the highest correlation with that cluster and lowest correlation to other clusters.



# Variable Clustering





# Weight of Evidences

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| Information Value | Predictive Power                  |
|-------------------|-----------------------------------|
| < 0.02            | Useless for prediction            |
| 0.02 to 0.1       | Weak predictor                    |
| 0.1 to 0.3        | Medium predictor                  |
| 0.3 to 0.5        | Strong predictor                  |
| > 0.5             | Suspicious or too good to be true |

Is high IV really bad ?

# Wrapper methods



| Technique                        | Explanation                                                                                                                                                                                                                                              |
|----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Forward Selection             | Start with a Null model. Add one variable at a time. Start with variable for which AIC is the lowest. Add next variable for which the AIC is lower. We can also use p value or AUC or <b>Marginal Information Value</b> or <b>Marginal Contributions</b> |
| 2. Backward Selection            | Start with a Full model. Eliminate one variable at a time. Start with the variable for which AIC is lowest. Eliminate next variable for which the AIC is lower.                                                                                          |
| 3. Stepwise Regression           | Suppose you add a variable through Forward regression. After adding a variable, you can eliminate an existing variable if p value is above 5% or VIF is above threshold.                                                                                 |
| 4. Sequential Forward            | At every step after you add new variables, you eliminate each of the existing variable and see if model performance can be improved. Also, model performance is checked on testing set.                                                                  |
| 5. Sequential Backward           | At every step after you eliminate variables, you add back each of the eliminated variable and see if model performance can be improved. Also, model performance is checked on testing set.                                                               |
| 6. Recursive Feature Elimination | Rank all the features based on absolute values of beta coefficients. Eliminate the least ranked feature. Repeat the process on balance variables.                                                                                                        |
| 7. Exhaustive Model Search       | Try all the possible combinations of models and choose the model which passes all the assumptions tests and has sufficient accuracy                                                                                                                      |

Note - We can also use some other techniques like Forced Variable selection or controlling selection sequence

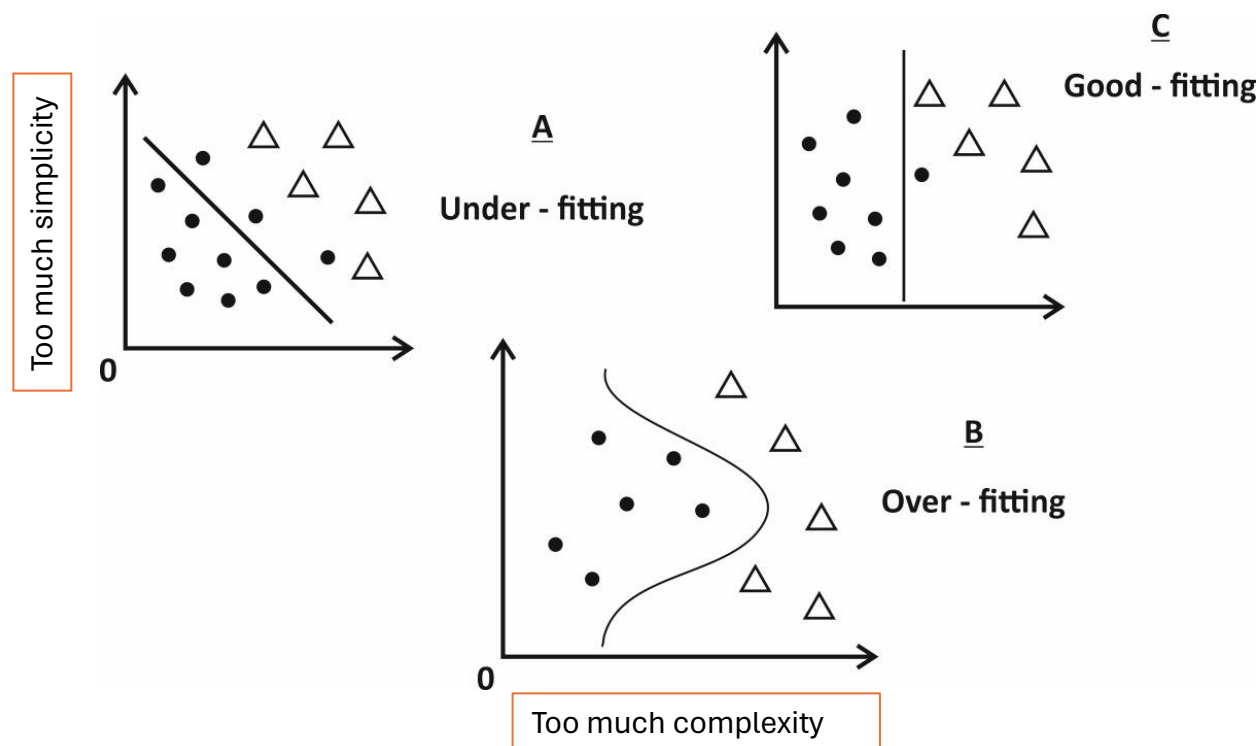
# Wrapper Methods

Which technique is the best ?

| C                             | D          | E        | F        |
|-------------------------------|------------|----------|----------|
|                               | AIC        |          |          |
| Forward Selection             | Null Model | Step 1   | Step 2   |
|                               | 8624.859   | 8026.281 | 5847.671 |
|                               | 8624.859   | 8425.242 | 6181.874 |
|                               | 8624.859   | 7226.04  | 5269.407 |
|                               | 8624.859   | 6341.862 | 6341.862 |
|                               | 8624.859   | 8380.29  | 6125.373 |
|                               | 8624.859   | 8206.623 | 6343.54  |
| Backward Selection            | Full Model | Step 1   | Step 2   |
|                               | 5011.788   | 5082.70  | 5080.702 |
|                               | 5011.788   | 5010.972 | 5008.978 |
|                               | 5011.788   | 5633.792 | 5631.843 |
|                               | 5011.788   | 6548.71  | 6908.918 |
|                               | 5011.788   | 5183.54  | 5181.564 |
|                               | 5011.788   | 5009.791 | 5009.79  |
| Recursive Feature Elimination | -2.87189   | 1        | -2.87122 |
|                               | -0.4781    | 5        | -0.47808 |
|                               | -0.09775   | 6        | -0.09787 |
|                               | -0.83377   | 4        | -0.83375 |
|                               | -0.99216   | 3        | -0.99236 |
|                               | -1.0037    | 2        | -1.00367 |
|                               | -0.00421   | 7        |          |

# Penalised Regression to reduce Overfitting

- **What leads to 'overfitting' and 'underfitting' in the data.**
- Overfitting is caused by taking extremely high no. of variables making the model complex.
- Underfitting is caused by taking extremely low no. of variables.



## # Reducing the problem of Overfitting

- Preventing the Algorithm from getting too complex requires estimating a penalty ( $\lambda$ ) for increase in complexity & proper data sampling achieved using validation ( k fold validation).
- When we add penalty terms to our regular regression model, it becomes penalized regression model.

## # Penalized Regression

- Penalized Regression is useful for reducing a large no. of features to a manageable set and for making good prediction in a variety of large data sets specially when the features (X's) are correlated.
- Penalised regression includes a constraints such that the regression coefficients are chosen to minimize the SSE + a penalty term that increases in size with the number of included features. So, in penalized regression a features must make a significant contribution to the model fit to offset penalty from including it.  $\therefore$  only the most imp features for explaining Y will remain in the penalised regression model.

# LASSO Regression

$$\lambda \sum |\beta|$$

# Ridge Regression

$$\lambda \sum (\beta)^2$$

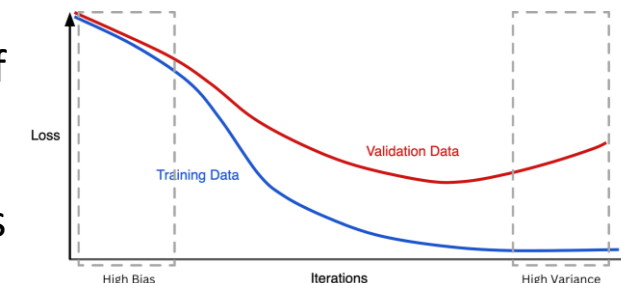
# Elastic Nets Regression

$$\lambda_1 \text{ and } \lambda_2$$

## Q. How to find $\lambda$ ? (Regularisation Parameter)

We choose that level of  $\lambda$ , for which the mean squared error of validation set is the lowest.

→ Finding out the MSE on validation set requires K – fold cross validation.



**Step 1:** For the first fold, on training data, take  $\lambda$  = say 0.1, run the penalized regression model and find out the Beta Coefficients.

**Step 2:** Based on estimated Beta Coefficients find out Y for the first fold validation set and thus the error terms, now you can calculate mean square error in the validation set.

**Step 3:** If K = 4, repeat the above 2 steps 3 times more and collect MSE, in validation fold 2, fold 3, fold 4. Take the average MSE.

**Step 4:** Repeat all the above 3 steps taking lambda = say 0.3.

**Step 5:** Plot the average MSE validation against possible  $\lambda$  (lambdas) and choose that  $\lambda$  which gives the lowest average MSE in the validation sets.

## Regression

Linear Regression

Logistic Regression

## Objective Function for Ridge

$$\text{SSE} + \lambda \sum (\beta)^2$$

$$-\text{Log Likelihood} + \lambda \sum (\beta)^2$$

$\min \sum (Y - XB)^2 + \lambda \sum \beta^2$  , if  $\lambda = 0$  , it becomes OLS. Increasing  $\lambda$  will drive the coefficients to zero.

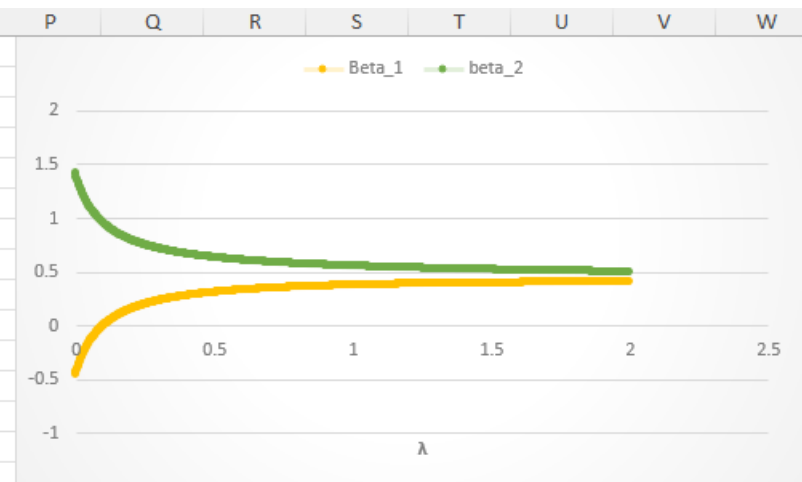
It's easy to see that in the above optimization, high values of beta coefficients are suppressed by use of the penalty factor. High values of beta denotes that the prediction is heavily following the learning set. This has issues of Overfitting the data. Ridge regression introduces bias in those beta coefficients and so the coefficients are not Unbiased anymore, however it reduces the variance of those estimates thereby improving generalization.

Analytical solution for Ridge estimators exist and is given by :

| Standardized X, Y |          |   |           |   |   |   |       | B = ( X' X + λ I ) <sup>-1</sup> X' Y |                |                | SSE      |
|-------------------|----------|---|-----------|---|---|---|-------|---------------------------------------|----------------|----------------|----------|
| X                 |          |   | Y         | I |   |   | λ     | β <sub>1</sub>                        | β <sub>2</sub> | β <sub>0</sub> |          |
| -1.59934          | -1.65778 | 1 | -1.666019 | 1 | 0 | 0 | 0     | -0.461689                             | 1.41254607     | -6.8835E-17    | 1.588993 |
| -1.42164          | -1.35491 | 1 | -1.222266 | 0 | 1 | 0 | 0.002 | -0.444168                             | 1.39497126     | -6.7192E-17    | 1.589057 |
| -1.24393          | -1.05205 | 1 | -1.074348 | 0 | 0 | 1 | 0.004 | -0.427291                             | 1.37804158     | -6.561E-17     | 1.589242 |
| -1.06623          | -0.90062 | 1 | -0.778513 |   |   |   | 0.006 | -0.411024                             | 1.3617221      | -6.4085E-17    | 1.589533 |
| -0.88852          | -0.90062 | 1 | -0.926431 |   |   |   | 0.008 | -0.395336                             | 1.34598038     | -6.2615E-17    | 1.589919 |
| -0.71082          | -0.90062 | 1 | -0.926431 |   |   |   | 0.01  | -0.380194                             | 1.3307862      | -6.1196E-17    | 1.590391 |
| -0.53311          | -0.74919 | 1 | -0.630596 |   |   |   | 0.012 | -0.365573                             | 1.31611145     | -5.9826E-17    | 1.590937 |
| -0.35541          | -0.44632 | 1 | -0.334761 |   |   |   | 0.014 | -0.351444                             | 1.3019299      | -5.8502E-17    | 1.591551 |
| -0.1777           | -0.14346 | 1 | -0.038926 |   |   |   | 0.016 | -0.337784                             | 1.28821704     | -5.7222E-17    | 1.592225 |
| 0                 | 0.00797  | 1 | 0.108992  |   |   |   | 0.018 | -0.32457                              | 1.27494998     | -5.5984E-17    | 1.592952 |
| 0.177705          | 0.159401 | 1 | -0.334761 |   |   |   | 0.02  | -0.31178                              | 1.26210729     | -5.4787E-17    | 1.593725 |
| 0.355409          | 0.462264 | 1 | 0.84858   |   |   |   | 0.022 | -0.299395                             | 1.24966888     | -5.3627E-17    | 1.59454  |
| 0.533114          | 0.462264 | 1 | 0.256909  |   |   |   | 0.024 | -0.287395                             | 1.23761593     | -5.2503E-17    | 1.595391 |
| 0.710819          | 0.765127 | 1 | 1.144415  |   |   |   | 0.026 | -0.275762                             | 1.22593075     | -5.1414E-17    | 1.596274 |
| 0.888523          | 0.916559 | 1 | 0.922538  |   |   |   | 0.028 | -0.264481                             | 1.21459674     | -5.0358E-17    | 1.597185 |
| 1.066228          | 1.06799  | 1 | 0.330868  |   |   |   | 0.03  | -0.253536                             | 1.20359824     | -4.9333E-17    | 1.598121 |
| 1.243933          | 1.370853 | 1 | 1.366291  |   |   |   | 0.032 | -0.242911                             | 1.19292057     | -4.8339E-17    | 1.599077 |
| 1.421637          | 1.370853 | 1 | 1.884002  |   |   |   | 0.034 | -0.232593                             | 1.18254983     | -4.7373E-17    | 1.600052 |
| 1.599342          | 1.522284 | 1 | 1.070456  |   |   |   | 0.036 | -0.222569                             | 1.17247297     | -4.6435E-17    | 1.601042 |
|                   |          |   |           |   |   |   | 0.038 | -0.212827                             | 1.16267762     | -4.5524E-17    | 1.602045 |
|                   |          |   |           |   |   |   | 0.04  | -0.203354                             | 1.15315213     | -4.4638E-17    | 1.603059 |
|                   |          |   |           |   |   |   | 0.042 | -0.19414                              | 1.14388546     | -4.3776E-17    | 1.604082 |
|                   |          |   |           |   |   |   | 0.044 | -0.185175                             | 1.13486718     | -4.2938E-17    | 1.605111 |
|                   |          |   |           |   |   |   | 0.046 | -0.176448                             | 1.12608741     | -4.2122E-17    | 1.606146 |
|                   |          |   |           |   |   |   | 0.048 | -0.16795                              | 1.11753678     | -4.1327E-17    | 1.607186 |
|                   |          |   |           |   |   |   | 0.05  | -0.159672                             | 1.10920642     | -4.0554E-17    | 1.608227 |
|                   |          |   |           |   |   |   | 0.052 | -0.151607                             | 1.10108788     | -3.98E-17      | 1.60927  |

Penalised Regression – Linear Regression

## Penalised Regression – Linear Regression



There are no statistical procedures to estimate  $\lambda$ .

- One way to estimate  $\lambda$  is to visually inspect the Ridge Trace plot which plots values of beta coefficients for various values of  $\lambda$ . Then the choice of  $\lambda$  is based on where the coefficients stabilize. In the diagram above  $\lambda = 0.4$  can be a candidate. The beta values are picked up for this value of  $\lambda$ .

- Another way to estimate will be to do a K-fold cross validation and choose  $\lambda$  that gives lowest SSE. But , this will be computationally intensive.



# Penalised Regression - Logistic



PEAKS2TAILS  
http://peaks2tails.com

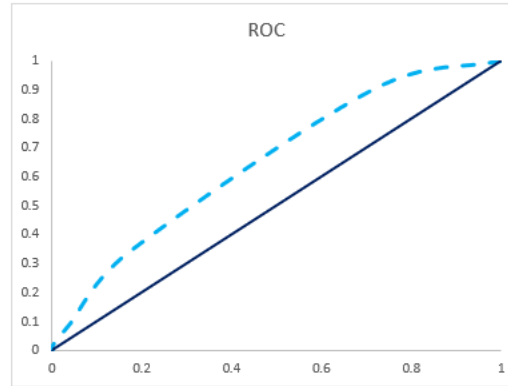


## Regression (Hyperparameter Tuning)

$\lambda =$   Loss **1422.67**

There are several hyper parameters.

1. We can choose ROC AUC ( threshold agnostic ) as performance measure and tune for the penalization factor  $\lambda$  on validation set
2. We can do regular logistic regression without regularization term and tune the threshold to optimize say recall or precision or F1-score etc.
3. We can do a grid search in the hyper parameter space ( norm ,  $\lambda$  ), train betas on training set and measure performance on validation set for each hyperparameter combination
4. If we are combining model selection and hyperparameter tuning using K cross fold, we must use nested CV scheme.

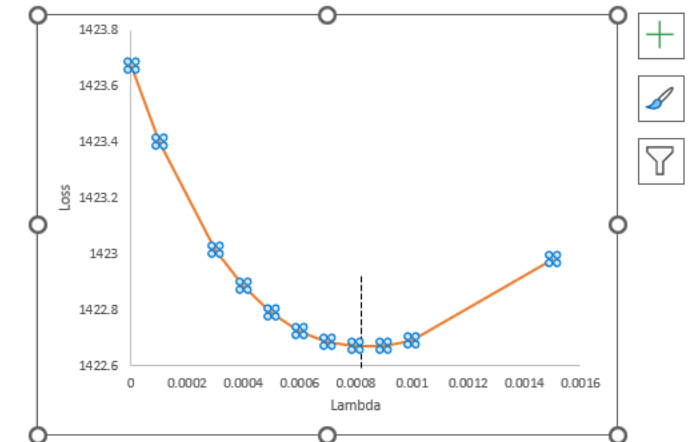


ROC AUC **0.63868**

$\lambda(\text{optimal})$  **0.008**

| $\lambda$ | Loss        |
|-----------|-------------|
| 0         | 1423.672214 |
| 0.0001    | 1423.399686 |
| 0.0003    | 1423.013064 |
| 0.0004    | 1422.883615 |
| 0.0005    | 1422.788694 |
| 0.0006    | 1422.723549 |
| 0.0007    | 1422.684168 |
| 0.0008    | 1422.667167 |
| 0.0009    | 1422.669701 |
| 0.001     | 1422.689329 |
| 0.0015    | 1422.980361 |

| p (cut-off)    | 0.00001   | 0.1     | 0.2     | 0.3     | 0.4     | 0.5     | 0.6         | 0.7         | 0.8     | 0.9     | 1       |
|----------------|-----------|---------|---------|---------|---------|---------|-------------|-------------|---------|---------|---------|
| TP             | 544       | 489     | 183     | 51      | 11      | 0       | 0           | 0           | 0       | 0       | 0       |
| FP             | 2904      | 2070    | 489     | 117     | 6       | 1       | 0           | 0           | 0       | 0       | 0       |
| TN             | 0         | 834     | 2415    | 2787    | 2898    | 2903    | 2904        | 2904        | 2904    | 2904    | 2904    |
| FN             | 0         | 55      | 361     | 493     | 533     | 544     | 544         | 544         | 544     | 544     | 544     |
| FPR            | 1         | 0.71281 | 0.16839 | 0.04029 | 0.00207 | 0.00034 | 0           | 0           | 0       | 0       | 0       |
| TPR            | 1         | 0.8989  | 0.3364  | 0.09375 | 0.02022 | 0       | 0           | 0           | 0       | 0       | 0       |
| Random Model   | 1         | 0.71281 | 0.16839 | 0.04029 | 0.00207 | 0.00034 | 0           | 0           | 0       | 0       | 0       |
| Accuracy       | 0.1577726 | 0.3837  | 0.75348 | 0.82309 | 0.84368 | 0.84194 | 0.842227378 | 0.842227378 | 0.84223 | 0.84223 | 0.84223 |
| Precision      | 0.1577726 | 0.19109 | 0.27232 | 0.30357 | 0.64706 | 0       | 0           | 0           | 0       | 0       | 0       |
| Recall         | 1         | 0.8989  | 0.3364  | 0.09375 | 0.02022 | 0       | 0           | 0           | 0       | 0       | 0       |
| F1_Score       | 0.2725451 | 0.31518 | 0.30099 | 0.14326 | 0.03922 | 0       | 0           | 0           | 0       | 0       | 0       |
| Trapezium Area | 0.2726722 | 0.33626 | 0.02755 | 0.00218 | 1.7E-05 | 0       | 0           | 0           | 0       | 0       | 0       |

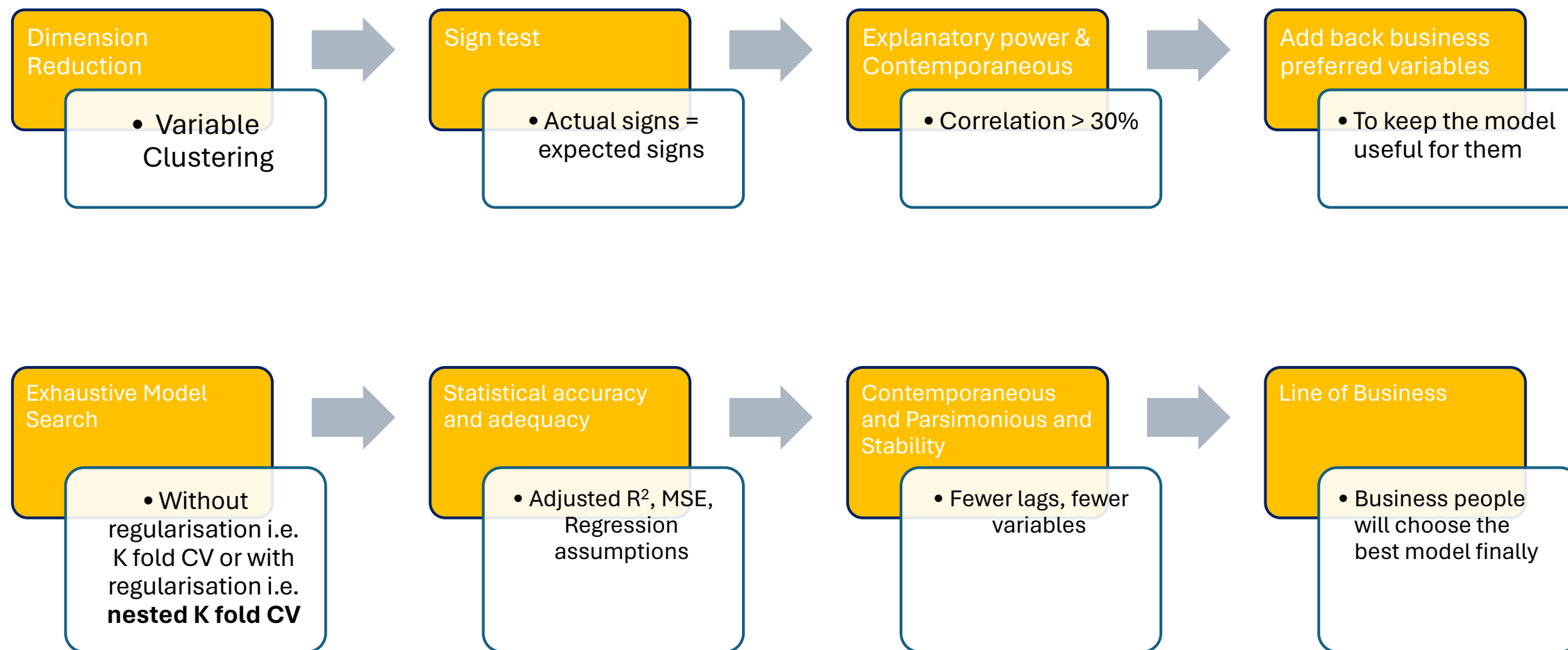


| #  | Annual_Inc | an_Purpo | an_Grad | an_Statu | P(Y=1)  | P(Y=0)  | LL           | Predicted | Predicted | Predicted | Predicted | Predicted | Predicted | Predicted | Predicted | Predicted |
|----|------------|----------|---------|----------|---------|---------|--------------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|
| 1  | -0.00397   | -0.0225  | 0.06246 | 0        | 0.15411 | 0.84589 | -0.167366295 | 1         | 1         | 0         | 0         | 0         | 0         | 0         | 0         | 0         |
| 3  | 0.3260051  | 0.01253  | -1.0414 | 1        | 0.07906 | 0.92094 | -2.537578867 | 1         | 0         | 0         | 0         | 0         | 0         | 0         | 0         | 0         |
| 21 | -0.066145  | 0.15855  | 0.06246 | 0        | 0.1649  | 0.8351  | -0.180203735 | 1         | 1         | 0         | 0         | 0         | 0         | 0         | 0         | 0         |
| 31 | -0.093484  | 0.72815  | 0.3625  | 0        | 0.27782 | 0.72218 | -0.325486714 | 1         | 1         | 1         | 0         | 0         | 0         | 0         | 0         | 0         |
| 35 | -0.00397   | 0.16009  | 0.3625  | 1        | 0.21573 | 0.78427 | -1.533741787 | 1         | 1         | 1         | 0         | 0         | 0         | 0         | 0         | 0         |
| 37 | -0.093484  | -0.3742  | 0.06246 | 0        | 0.1167  | 0.8833  | -0.124086353 | 1         | 1         | 0         | 0         | 0         | 0         | 0         | 0         | 0         |
| 45 | 0.3260051  | 0.15855  | -0.2444 | 0        | 0.16758 | 0.83242 | -0.183417185 | 1         | 1         | 0         | 0         | 0         | 0         | 0         | 0         | 0         |
| 50 | -0.066145  | -0.0225  | 0.3625  | 1        | 0.18697 | 0.81303 | -1.67681779  | 1         | 1         | 0         | 0         | 0         | 0         | 0         | 0         | 0         |

< > Master Validation\_nonregularized Validation\_Regularized cost\_Tuning +



## Regression Pipeline

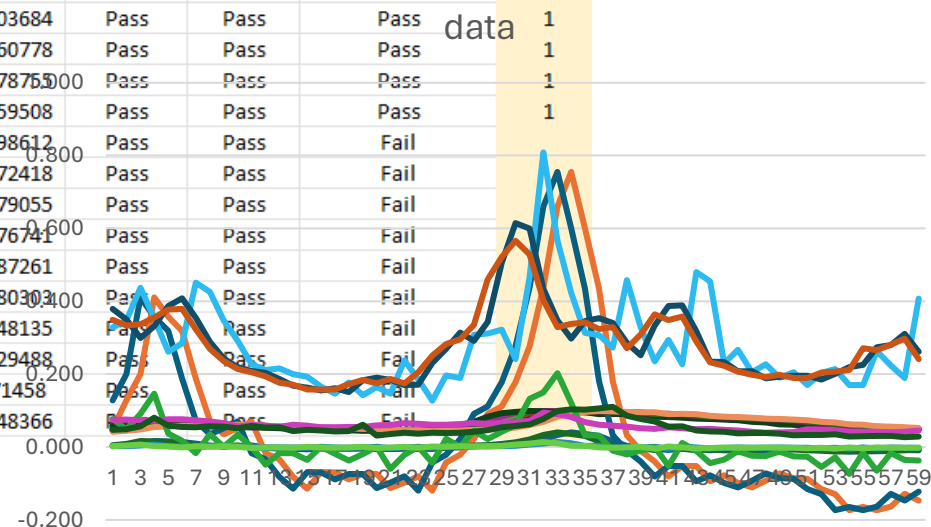


## Variable transformation list

\_log  
 \_lag\_1  
 \_lag\_2  
 \_lag\_3  
 \_lag\_4  
 \_lead\_1  
 \_lead\_2  
 \_lead\_3  
 \_lead\_4  
 \_qoq\_diff  
 \_qoq\_diff\_lag\_1  
 \_qoq\_diff\_lag\_2  
 \_qoq\_diff\_lag\_3  
 \_qoq\_diff\_lag\_4  
 \_yoy\_diff  
 \_yoy\_diff\_lag\_1  
 \_yoy\_diff\_lag\_2  
 \_yoy\_diff\_lag\_3  
 \_yoy\_diff\_lag\_4  
 \_qoq\_log\_growth  
 \_qoq\_log\_growth\_lag\_1  
 \_qoq\_log\_growth\_lag\_2  
 \_qoq\_log\_growth\_lag\_3  
 \_qoq\_log\_growth\_lag\_4  
 \_qoq\_simple\_growth  
 \_qoq\_simple\_growth\_lag\_1  
 \_qoq\_simple\_growth\_lag\_2  
 \_qoq\_simple\_growth\_lag\_3  
 \_qoq\_simple\_growth\_lag\_4  
 \_yoy\_log\_growth  
 \_yoy\_log\_growth\_lag\_1  
 \_yoy\_log\_growth\_lag\_2  
 \_yoy\_log\_growth\_lag\_3  
 \_yoy\_log\_growth\_lag\_4  
 \_yoy\_simple\_growth  
 \_yoy\_simple\_growth\_lag\_1  
 \_yoy\_simple\_growth\_lag\_2  
 \_yoy\_simple\_growth\_lag\_3  
 \_yoy\_simple\_growth\_lag\_4  
 \_qqma2\_leading  
 \_qqma3\_leading  
 \_qqma4\_leading  
 \_qqma2\_lagging  
 \_qqma3\_lagging  
 \_qqma4\_lagging



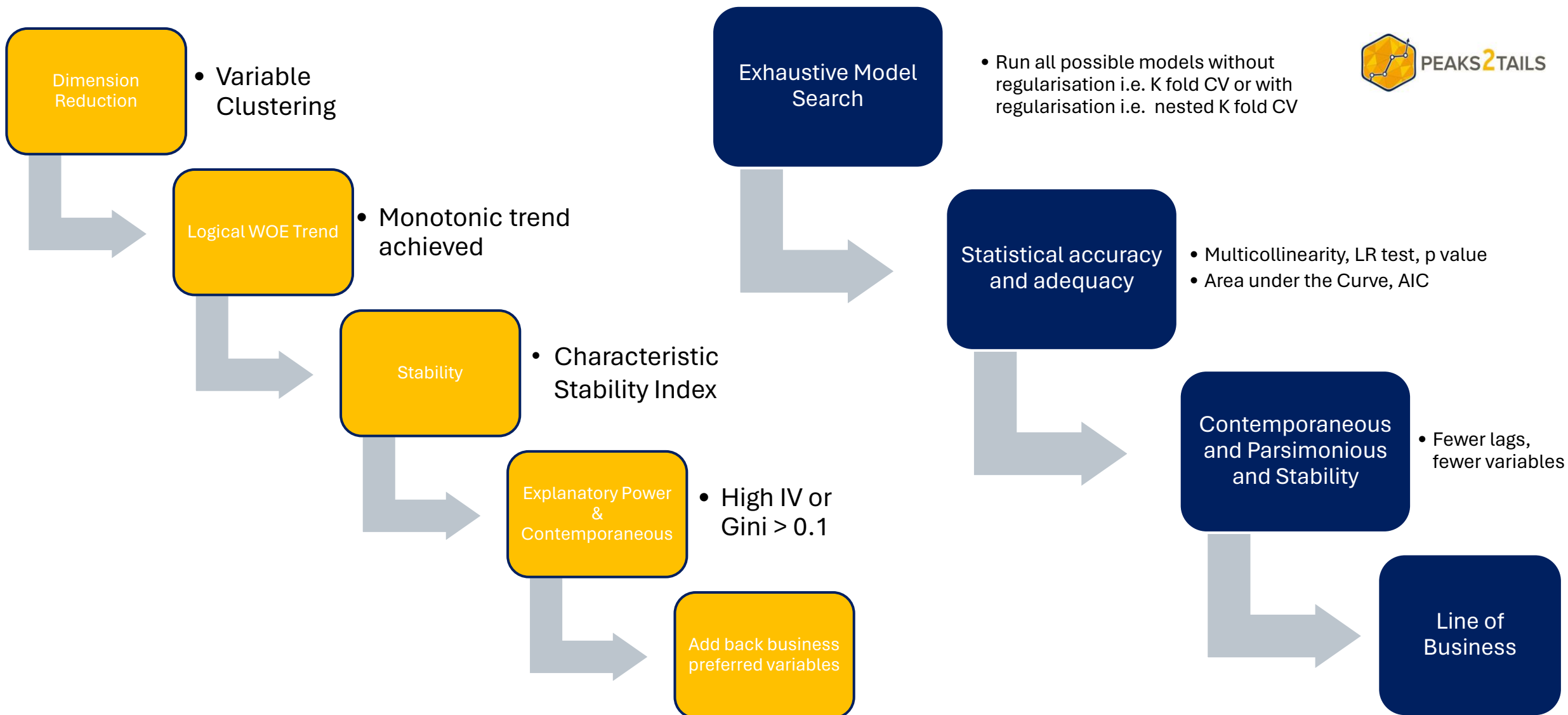
| X                                    | Y    | Correlation | Significance | Sign Test | Contemporaneous | Include |
|--------------------------------------|------|-------------|--------------|-----------|-----------------|---------|
| Unemployment_yoy_diff_lag_1          | COCC | 0.851628179 | Pass         | Pass      | Pass            | 1       |
| Unemployment_yoy_simple_growth_lag_1 | COCC | 0.815835505 | Pass         | Pass      | Pass            | 1       |
| Unemployment_yoy_diff                | COCC | 0.730045658 | Pass         | Pass      | Pass            | 1       |
| Unemployment_lead_2                  | COCC | 0.72371253  | Pass         | Pass      | Pass            | 1       |
| Unemployment_lead_3                  | COCC | 0.71934391  | Pass         | Pass      | Pass            | 1       |
| Unemployment_qqma4_leading           | COCC | 0.708938319 | Pass         | Pass      | Pass            | 1       |
| Unemployment_lead_1                  | COCC | 0.704786384 | Pass         | Pass      | Pass            | 1       |
| Unemployment_qqma3_leading           | COCC | 0.698916101 | Pass         | Pass      | Pass            | 1       |
| Unemployment_lead_4                  | COCC | 0.692612177 | Pass         | Pass      | Pass            | 1       |
| Unemployment_yoy_simple_growth       | COCC | 0.687725973 | Pass         | Pass      | Pass            | 1       |
| Unemployment_qqma2_leading           | COCC | 0.683319395 | Pass         | Pass      | Pass            | 1       |
| Unemployment                         | COCC | 0.657241394 | Pass         | Pass      | Pass            | 1       |
| Unemployment_qoq_diff_lag_1          | COCC | 0.600791918 | Pass         | Pass      | Pass            | 1       |
| Unemployment_lag_1                   | COCC | 0.576505577 | Pass         | Pass      | Pass            | 1       |
| Unemployment_qoq_simple_growth_lag_1 | COCC | 0.567284802 | Pass         | Pass      | Pass            | 1       |
| Market_Volatility_lag_1              | COCC | 0.481355363 | Pass         | Pass      | Pass            | 1       |
| BBB_corporate_yield_lag_1            | COCC | 0.472303684 | Pass         | Pass      | Pass            | 1       |
| Unemployment_qoq_diff                | COCC | 0.410060778 | Pass         | Pass      | Pass            | 1       |
| Market_Volatility_qqma3_leading      | COCC | 0.401678755 | Pass         | Pass      | Pass            | 1       |
| Market_Volatility_qqma4_leading      | COCC | 0.400159508 | Pass         | Pass      | Pass            | 1       |
| Unemployment_yoy_diff_lag_3          | COCC | 0.903198612 | Pass         | Pass      | Fail            |         |
| Unemployment_yoy_diff_lag_2          | COCC | 0.900272418 | Pass         | Pass      | Fail            |         |
| Unemployment_yoy_simple_growth_lag_3 | COCC | 0.884279055 | Pass         | Pass      | Fail            |         |
| Unemployment_yoy_simple_growth_lag_2 | COCC | 0.870076741 | Pass         | Pass      | Fail            |         |
| Unemployment_yoy_diff_lag_4          | COCC | 0.853287261 | Pass         | Pass      | Fail            |         |
| Unemployment_yoy_simple_growth_lag_4 | COCC | 0.845380301 | Pass         | Pass      | Fail            |         |
| Unemployment_qoq_diff_lag_4          | COCC | 0.825848135 | Pass         | Pass      | Fail            |         |
| Unemployment_qoq_simple_growth_lag_4 | COCC | 0.807029488 | Pass         | Pass      | Fail            |         |
| Unemployment_qoq_diff_lag_3          | COCC | 0.78471458  | Pass         | Pass      | Fail            |         |
| Unemployment_qoq_simple_growth_lag_3 | COCC | 0.757448366 | Pass         | Pass      | Fail            |         |



|                                      | Unemployment_yoy_diff_lag_1 | Unemployment_yoy_simple_growth_lag_1 | Unemployment_yoy_diff | Unemployment_lead_2 | Unemployment_lead_3 | Unemployment_qqma4_leading | Unemployment_lead_1 | Unemployment_qqma3_leading | Unemployment_lead_4 | Unemployment_yoy_simple_growth | Unemployment_qqma2_leading | Unemployment | Unemployment_qoq_diff_lag_1 | Unemployment_qoq_diff_lag_1 | Unemployment_qoq_simple_growth_lag_1 | Market_Volatility_lag_1 | BBB_corporate_yield_lag_1 | Unemployment_qoq_diff | Market_Volatility_qqma3_leading | Market_Volatility_qqma4_leading |
|--------------------------------------|-----------------------------|--------------------------------------|-----------------------|---------------------|---------------------|----------------------------|---------------------|----------------------------|---------------------|--------------------------------|----------------------------|--------------|-----------------------------|-----------------------------|--------------------------------------|-------------------------|---------------------------|-----------------------|---------------------------------|---------------------------------|
| Unemployment_yoy_diff_lag_1          | 1                           | 0.98472452                           | 0.93812204            | 0.54879409          | 0.58526774          | 0.51429639                 | 0.48547186          | 0.48151531                 | 0.60015447          | 0.89062839                     | 0.4409975                  | 0.39204652   | 0.81376397                  | 0.26394105                  | 0.75967571                           | 0.54900218              | 0.77486623                | 0.60858016            | 0.511459                        | 0.51777967                      |
| Unemployment_yoy_simple_growth_lag_1 | 0.98472452                  | 1                                    | 0.95357993            | 0.56061285          | 0.597288            | 0.52402551                 | 0.49551858          | 0.49030328                 | 0.60979024          | 0.93053859                     | 0.44814302                 | 0.39623011   | 0.84699984                  | 0.26047804                  | 0.80726435                           | 0.59252346              | 0.78588403                | 0.64572201            | 0.5438797                       | 0.55204702                      |
| Unemployment_yoy_diff                | 0.93812204                  | 0.95357993                           | 1                     | 0.49624487          | 0.55567199          | 0.443296                   | 0.40510817          | 0.3977117                  | 0.59112525          | 0.9847848                      | 0.34272758                 | 0.27701839   | 0.89815399                  | 0.10693516                  | 0.86805748                           | 0.64429226              | 0.85141383                | 0.81833732            | 0.58199555                      | 0.58863205                      |
| Unemployment_lead_2                  | 0.54879409                  | 0.56061285                           | 0.49624487            | 1                   | 0.98046294          | 0.99359206                 | 0.97891564          | 0.9811081                  | 0.93281476          | 0.50659796                     | 0.95774674                 | 0.92644773   | 0.41028536                  | 0.84877044                  | 0.40587627                           | 0.56344525              | 0.19902836                | 0.33977704            | 0.51012298                      | 0.48090743                      |
| Unemployment_lead_3                  | 0.58526774                  | 0.597288                             | 0.55567199            | 0.98046294          | 1                   | 0.96222936                 | 0.92992846          | 0.93305339                 | 0.98074615          | 0.56789916                     | 0.89608995                 | 0.85284938   | 0.4678622                   | 0.75791475                  | 0.47143754                           | 0.59099056              | 0.27693365                | 0.42712648            | 0.61778216                      | 0.60408134                      |
| Unemployment_qqma4_leading           | 0.51429639                  | 0.52402551                           | 0.443296              | 0.99359206          | 0.96222936          | 1                          | 0.99403752          | 0.99574644                 | 0.90399821          | 0.44939407                     | 0.9827186                  | 0.96094918   | 0.3525121                   | 0.8980552                   | 0.34213593                           | 0.5281882               | 0.13458932                | 0.26616185            | 0.4374593                       | 0.4093512                       |
| Unemployment_lead_1                  | 0.48547186                  | 0.49551858                           | 0.40510817            | 0.97891564          | 0.92992846          | 0.99403752                 | 1                   | 0.99879941                 | 0.85814248          | 0.40960099                     | 0.99460796                 | 0.97860619   | 0.31519144                  | 0.9262848                   | 0.30069548                           | 0.50908238              | 0.0947787                 | 0.21381237            | 0.36630163                      | 0.3315497                       |
| Unemployment_qqma3_leading           | 0.48151531                  | 0.49030328                           | 0.3977117             | 0.9811081           | 0.93305339          | 0.99574644                 | 0.99879941          | 1                          | 0.86262296          | 0.40163156                     | 0.99529477                 | 0.98116284   | 0.30746789                  | 0.93018407                  | 0.2925468                            | 0.49793514              | 0.08412193                | 0.20715115            | 0.36897876                      | 0.33647357                      |
| Unemployment_lead_4                  | 0.60015447                  | 0.60979024                           | 0.59112525            | 0.93281476          | 0.98074615          | 0.90399821                 | 0.85814248          | 0.86262296                 | 1                   | 0.60372291                     | 0.81524699                 | 0.76386317   | 0.49670317                  | 0.6583495                   | 0.50814568                           | 0.58991591              | 0.3365682                 | 0.48253554            | 0.68347974                      | 0.69559637                      |
| Unemployment_yoy_simple_growth       | 0.89062839                  | 0.93053859                           | 0.9847848             | 0.50659796          | 0.56789916          | 0.44939407                 | 0.40960099          | 0.40163156                 | 0.60372291          | 1                              | 0.34336988                 | 0.27382411   | 0.91041642                  | 0.0968698                   | 0.90470605                           | 0.68735046              | 0.84652487                | 0.85200754            | 0.61994185                      | 0.62901321                      |
| Unemployment_qqma2_leading           | 0.4409975                   | 0.44814302                           | 0.34272758            | 0.95774674          | 0.89608995          | 0.9827186                  | 0.99460796          | 0.99529477                 | 0.81524699          | 0.34336988                     | 1                          | 0.99466629   | 0.25161651                  | 0.95778152                  | 0.23164194                           | 0.45807053              | 0.02538003                | 0.13779162            | 0.29306025                      | 0.25936797                      |
| Unemployment                         | 0.39204652                  | 0.39623011                           | 0.27701839            | 0.92644773          | 0.85284938          | 0.96094918                 | 0.97860619          | 0.98116284                 | 0.76386317          | 0.27382411                     | 0.99466629                 | 1            | 0.18569403                  | 0.97886289                  | 0.16048412                           | 0.40243481              | -0.0439147                | 0.06070815            | 0.21708029                      | 0.18480231                      |
| Unemployment_qoq_diff_lag_1          | 0.81376397                  | 0.84699984                           | 0.89815399            | 0.41028536          | 0.4678622           | 0.3525121                  | 0.31519144          | 0.30746789                 | 0.49670317          | 0.91041642                     | 0.25161651                 | 0.18569403   | 1                           | 0.03296806                  | 0.97791985                           | 0.60705502              | 0.78400154                | 0.73750149            | 0.51669507                      | 0.53762285                      |
| Unemployment_lag_1                   | 0.26394105                  | 0.26047804                           | 0.10693516            | 0.84877044          | 0.75791475          | 0.8980552                  | 0.9262848           | 0.93018407                 | 0.6583495           | 0.0968698                      | 0.95778152                 | 0.97886289   | 0.03296806                  | 1                           | 0.00235295                           | 0.26523633              | -0.2007261                | -0.1447154            | 0.08832415                      | 0.05952938                      |
| Unemployment_qoq_simple_growth_lag_1 | 0.75967571                  | 0.80726435                           | 0.86805748            | 0.40587627          | 0.47143754          | 0.34213593                 | 0.30069548          | 0.2925468                  | 0.50814568          | 0.90470605                     | 0.23164194                 | 0.16048412   | 0.97791985                  | 0.00235295                  | 1                                    | 0.64091697              | 0.78011419                | 0.76495204            | 0.57566067                      | 0.59434824                      |
| Market_Volatility_lag_1              | 0.54900218                  | 0.59252346                           | 0.64429226            | 0.56344525          | 0.59099056          | 0.5281882                  | 0.50908238          | 0.49793514                 | 0.58991591          | 0.68735046                     | 0.45807053                 | 0.40243481   | 0.60705502                  | 0.26523633                  | 0.64091697                           | 1                       | 0.58697487                | 0.65251789            | 0.50881002                      | 0.48157567                      |
| BBB_corporate_yield_lag_1            | 0.77486623                  | 0.78588403                           | 0.85141383            | 0.19902836          | 0.27693365          | 0.13458932                 | 0.0947787           | 0.08412193                 | 0.3365682           | 0.84652487                     | 0.02538003                 | -0.0439147   | 0.78400154                  | -0.2007261                  | 0.78011419                           | 0.58697487              | 1                         | 0.76718786            | 0.50373262                      | 0.50562303                      |
| Unemployment_qoq_diff                | 0.60858016                  | 0.64572201                           | 0.81833732            | 0.33977704          | 0.42712648          | 0.26616185                 | 0.21381237          | 0.20715115                 | 0.48253554          | 0.85200754                     | 0.13779162                 | 0.06070815   | 0.73750149                  | -0.1447154                  | 0.76495204                           | 0.65251789              | 0.76718786                | 1                     | 0.61918347                      | 0.60355386                      |
| Market_Volatility_qqma3_leading      | 0.511459                    | 0.5438797                            | 0.58199555            | 0.51012298          | 0.61778216          | 0.4374593                  | 0.36630163          | 0.36897876                 | 0.68347974          | 0.61994185                     | 0.29306025                 | 0.21708029   | 0.51669507                  | 0.08832415                  | 0.57566067                           | 0.50881002              | 0.50373262                | 0.61918347            | 1                               | 0.96130612                      |
| Market_Volatility_qqma4_leading      | 0.51777967                  | 0.55204702                           | 0.58863205            | 0.48090743          | 0.60408134          | 0.4093512                  | 0.3315497           | 0.33647357                 | 0.69559637          | 0.62901321                     | 0.25936797                 | 0.18480231   | 0.53762285                  | 0.05952938                  | 0.59434824                           | 0.48157567              | 0.50562303                | 0.60355386            | 0.96130612                      | 1                               |

Is Correlation sufficient to detect Multicollinearity ?





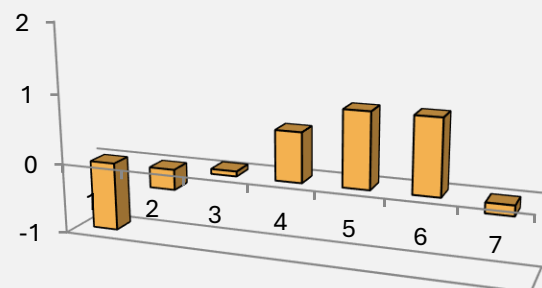
Classification  
Pipeline



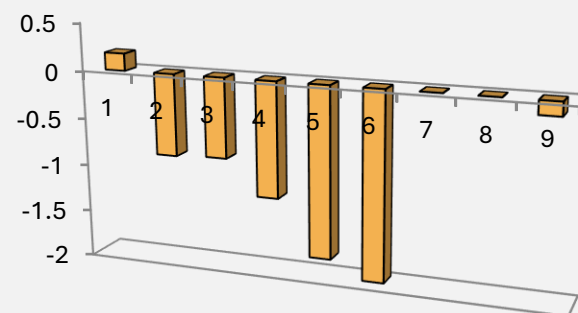
# What qualitative factors do business consider?

| Characteristics                 | Examples                                                                          |
|---------------------------------|-----------------------------------------------------------------------------------|
| I - Implementable               | LTV_time if no objective way of getting valuations timely                         |
| M - Manipulative                | Variables based on self-reported income                                           |
| P – Policy or Legal Constraints | Utilizing religion as a risk driver, Alternative data                             |
| O - Objective                   | Length of employment may have subjective interpretations (full time or part time) |
| R - Recognisable                | Variable like social media activity level not related to credit risk              |
| T - Transparency                | Calculation methodology of variable should be clear                               |
| A - Available                   | Data with high missing values like investment portfolio value                     |
| N - Necessary                   | Variables with low statistical performance but very important                     |
| T - Tangible                    | Intended use of loan although important but non- tangible                         |

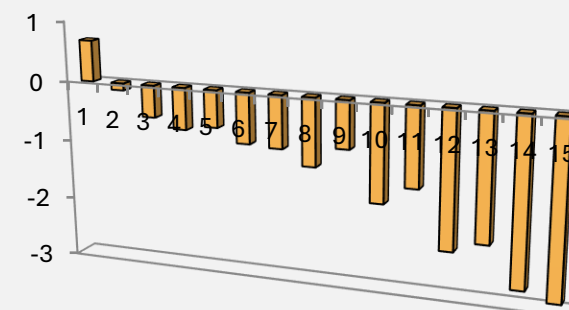
bureau\_score



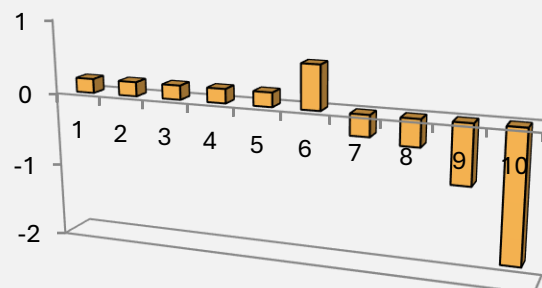
num\_ccj



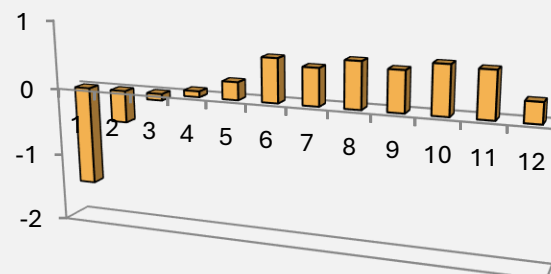
max\_arrears\_12m



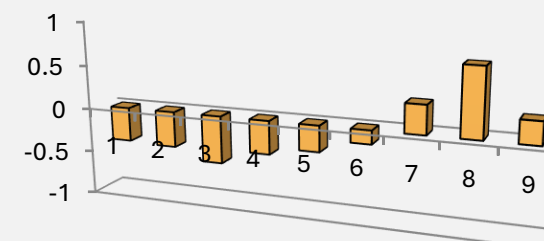
cc\_util



annual\_income



months\_since\_recent\_cc\_delinq



## Equal Frequency Bins

|    | A       | B            | C            | D                  | E           | F       | G             | H        | I         | J        | K        | L          | M          | N      | O      | P           | Q |
|----|---------|--------------|--------------|--------------------|-------------|---------|---------------|----------|-----------|----------|----------|------------|------------|--------|--------|-------------|---|
| 1  | WARNING |              |              |                    |             |         |               |          |           |          |          |            |            |        |        |             |   |
| 2  |         |              | bureau_score | num_ccj            | x_arrears_1 | cc_util | annual_income | ce_recer | mp_length |          |          |            |            |        |        |             |   |
| 3  |         |              | 6            | 8                  | 15          | 10      | 12            | 9        | 10        |          |          |            |            |        |        |             |   |
| 4  |         | Train        | num          | num                | num         | num     | num           | num      | cat       |          |          |            |            |        |        |             |   |
| 5  |         | 15           |              |                    |             |         |               |          |           |          |          |            |            |        |        |             |   |
| 6  |         |              |              |                    |             |         |               |          |           |          |          |            |            |        |        |             |   |
| 7  |         | bureau_score | Bins         | Description        | n           | goods   | bads          | %goods   | %bads     | woe      | IV       | UL         | LL         | A%     | E%     | CSI         |   |
| 8  |         |              | -Infinity    | $x < 275$          | 3466        | 3016    | 450           | 0.1537   | 0.407609  | -0.97522 | 0.247605 | < 275      | < 275      | 16.72% | 17.45% | 0.000307345 |   |
| 9  |         |              | 275          | $275 \leq x < 345$ | 3477        | 3238    | 239           | 0.165    | 0.216486  | -0.27141 | 0.013966 | < 345      | $\geq 275$ | 16.78% | 16.52% | 3.90356E-05 |   |
| 10 |         |              | 345          | $345 \leq x < 408$ | 3450        | 3281    | 169           | 0.1672   | 0.15308   | 0.088344 | 0.001249 | < 408      | $\geq 345$ | 16.65% | 16.60% | 1.35446E-06 |   |
| 11 |         |              | 408          | $408 \leq x < 464$ | 3460        | 3367    | 93            | 0.1716   | 0.084239  | 0.711517 | 0.06216  | < 464      | $\geq 408$ | 16.69% | 16.14% | 0.000190331 |   |
| 12 |         |              | 464          | $464 \leq x < 510$ | 3451        | 3379    | 72            | 0.1722   | 0.065217  | 0.971008 | 0.103894 | < 510      | $\geq 464$ | 16.65% | 15.98% | 0.000275092 |   |
| 13 |         |              | 510          | $x \geq 510$       | 3073        | 3011    | 62            | 0.1535   | 0.056159  | 1.005233 | 0.097808 | $\geq 510$ | $\geq 510$ | 14.83% | 15.67% | 0.000468494 |   |
| 14 |         |              | NA           | $x = \text{"NA"}$  | 348         | 329     | 19            | 0.0168   | 0.01721   | -0.02604 | 1.15E-05 | NA         | NA         | 1.68%  | 1.64%  | 8.94032E-06 |   |
| 15 |         |              |              |                    | 20725       | 19621   | 1104          |          |           |          | 0.526694 |            |            |        |        | 0.001290592 |   |
| 16 |         |              |              |                    |             |         |               |          |           |          |          |            |            |        |        |             |   |

Generate Bins

Clear Bins

## Monotonic Bins

|    |  | bureau_score | Bins      | Description        | n     | goods | bads | %goods      | %bads       | woe      | IV       | UL         | LL         | A%     | E%     | CSI         |
|----|--|--------------|-----------|--------------------|-------|-------|------|-------------|-------------|----------|----------|------------|------------|--------|--------|-------------|
| 7  |  |              | -Infinity | $x < 113$          | 121   | 85    | 36   | 0.004332093 | 0.032608696 | -2.01853 | 0.057077 | < 113      | < 113      | 0.58%  | 0.54%  | 3.35234E-05 |
| 8  |  |              | 113       | $113 \leq x < 256$ | 2557  | 2207  | 350  | 0.112481525 | 0.317028986 | -1.0362  | 0.211953 | < 256      | $\geq 113$ | 12.34% | 12.47% | 1.38105E-05 |
| 9  |  |              | 256       | $256 \leq x < 322$ | 3083  | 2842  | 241  | 0.144844809 | 0.218297101 | -0.41019 | 0.03013  | < 322      | $\geq 256$ | 14.88% | 15.06% | 2.14713E-05 |
| 10 |  |              | 322       | $322 \leq x < 392$ | 3694  | 3492  | 202  | 0.17797258  | 0.182971014 | -0.0277  | 0.000138 | < 392      | $\geq 322$ | 17.82% | 17.76% | 2.50029E-06 |
| 11 |  |              | 392       | $x \geq 392$       | 10922 | 10666 | 256  | 0.543601244 | 0.231884058 | 0.851978 | 0.265576 | $\geq 392$ | $\geq 392$ | 52.70% | 52.54% | 4.95794E-06 |
| 12 |  |              | NA        | $x = \text{"NA"}$  | 348   | 329   | 19   | 0.016767749 | 0.017210145 | -0.02604 | 1.15E-05 | NA         | NA         | 1.68%  | 1.64%  | 8.94032E-06 |
| 13 |  |              |           |                    | 20725 | 19621 | 1104 |             |             |          | 0.564886 |            |            |        |        | 8.52039E-05 |
| 14 |  |              |           |                    |       |       |      |             |             |          |          |            |            |        |        |             |
| 15 |  |              |           |                    |       |       |      |             |             |          |          |            |            |        |        |             |



# Exhaustive Model Search

| Models  | LL1        | k | AIC         | AUC         | AR          | R-Sq (L) | LR test |
|---------|------------|---|-------------|-------------|-------------|----------|---------|
| 1       | -4969.5238 | 1 | 9943.047641 | 0.698191698 | 0.396383396 | 0.06892  | PASS    |
| 2       | -5216.3622 | 1 | 10436.72438 | 0.574987358 | 0.149974717 | 0.02267  | PASS    |
| 3       | -4509.5391 | 1 | 9023.078122 | 0.6981917   | 0.457893684 | 0.1551   | PASS    |
| 4       | -3949.1053 | 1 | 7902.210679 | 0.6981917   | 0.7140637   | 0.2601   | PASS    |
| 5       | -4927.3658 | 1 | 9858.731612 | 0.6981917   | 0.357171579 | 0.07681  | PASS    |
| 6       | -5191.5116 | 1 | 10387.02329 | 0.632261779 | 0.264523559 | 0.02732  | PASS    |
| 7       | -5069.858  | 1 | 10143.71596 | 0.6981917   | 0.304105953 | 0.05012  | PASS    |
| 1, 2    | -4948.7339 | 2 | 9903.467843 | 0.6981917   | 0.410079557 | 0.07281  | PASS    |
| 1, 3    | -4441.0141 | 2 | 8888.028119 | 0.6981917   | 0.550395663 | 0.16794  | PASS    |
| 1, 4    | -3635.9622 | 2 | 7277.924363 | 0.6981917   | 0.792727813 | 0.31877  | PASS    |
| 1, 5    | -4573.3658 | 2 | 9152.731679 | 0.6981917   | 0.543971916 | 0.14314  | PASS    |
| 1, 6    | -4825.5464 | 2 | 9657.092817 | 0.736289839 | 0.472579677 | 0.09589  | PASS    |
| 1, 7    | -4709.7088 | 2 | 9425.417533 | 0.6981917   | 0.529199889 | 0.11759  | PASS    |
| 2, 3    | -4500.3046 | 2 | 9006.609274 | 0.743621183 | 0.487242367 | 0.15683  | PASS    |
| 2, 4    | -3851.563  | 2 | 7709.125986 | 0.871133379 | 0.742266758 | 0.27838  | PASS    |
| 2, 5    | -4819.8973 | 2 | 9645.794555 | 0.709295763 | 0.418591526 | 0.09695  | PASS    |
| 2, 6    | -5075.1559 | 2 | 10156.31174 | 0.6981917   | 0.33613447  | 0.04912  | PASS    |
| 2, 7    | -4952.2805 | 2 | 9910.560922 | 0.6981917   | 0.382506433 | 0.07215  | PASS    |
| 3, 4    | -3312.5496 | 2 | 6631.099223 | 0.912182586 | 0.824365173 | 0.37936  | PASS    |
| 3, 5    | -4170.5956 | 2 | 8347.191103 | 0.804806424 | 0.609612848 | 0.2186   | PASS    |
| 3, 6    | -4381.9167 | 2 | 8769.833443 | 0.779801015 | 0.559602031 | 0.17901  | PASS    |
| 3, 7    | -4268.2193 | 2 | 8542.438697 | 0.797201738 | 0.594403476 | 0.20031  | PASS    |
| 4, 5    | -3613.8962 | 2 | 7233.792486 | 0.894434979 | 0.788869957 | 0.3229   | PASS    |
| 4, 6    | -3816.507  | 2 | 7639.01393  | 0.880572793 | 0.761145586 | 0.28494  | PASS    |
| 4, 7    | -3949.038  | 2 | 7904.076038 | 0.855667552 | 0.711335105 | 0.26011  | PASS    |
| 5, 6    | -4923.7779 | 2 | 9853.55581  | 0.6981917   | 0.3963834   | 0.07749  | PASS    |
| 5, 7    | -4677.3059 | 2 | 9360.61172  | 0.6981917   | 0.3963834   | 0.12367  | PASS    |
| 6, 7    | -4923.989  | 2 | 9853.977977 | 0.719314032 | 0.3963834   | 0.07745  | PASS    |
| 1, 2, 3 | -4440.0038 | 3 | 8888.007595 | 0.6981917   | 0.552643926 | 0.16813  | PASS    |
| 1, 2, 4 | -3621.5509 | 3 | 7251.101784 | 0.6981917   | 0.795519976 | 0.32147  | PASS    |
| 1, 2, 5 | -4557.0792 | 3 | 9122.158404 | 0.6981917   | 0.3963834   | 0.14619  | PASS    |
| 1, 2, 6 | -4806.538  | 3 | 9621.076039 | 0.6981917   | 0.480567818 | 0.09945  | PASS    |
| 1, 2, 7 | -4689.6439 | 3 | 9387.287897 | 0.6981917   | 0.536499106 | 0.12135  | PASS    |
| 1, 3, 4 | -3250.6149 | 3 | 6509.22972  | 0.6981917   | 0.841474396 | 0.39097  | PASS    |
| 1, 3, 5 | -4100.7305 | 3 | 8209.461011 | 0.6981917   | 0.650955663 | 0.23169  | PASS    |
| 1, 3, 6 | -4313.6416 | 3 | 8635.283111 | 0.6981917   | 0.3963834   | 0.1918   | PASS    |
| 1, 3, 7 | -4198.9697 | 3 | 8405.939471 | 0.6981917   | 0.64121075  | 0.21329  | PASS    |

# Important Thresholds

| Metric                               | Threshold    |
|--------------------------------------|--------------|
| Characteristic Stability Index       | < 0.1        |
| Gini                                 | > 0.1        |
| Information Value                    | > 0.02       |
| Multicollinearity correlation cutoff | < 0.5 to 0.7 |
| Multicollinearity IV cut off         | < 2 to 3     |
| Model AUC                            | > 0.7        |
| Model Gini                           | > 0.4 to 0.5 |
| Model PSI                            | <0.1         |

Gini or IV which is more important ?